Stage

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Projektiranje autonomnih sustava| Regional Center of Excellence for Robotic Technology

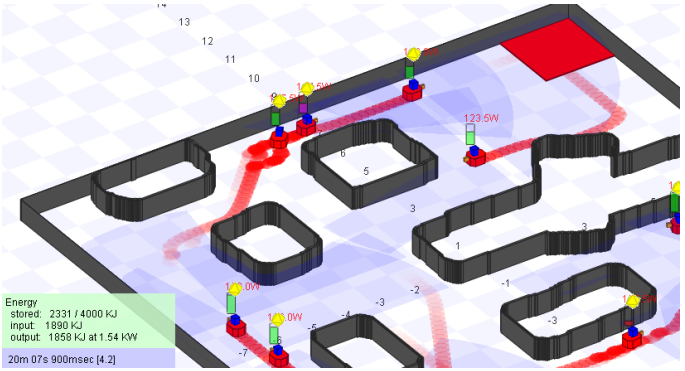
Sadržaj

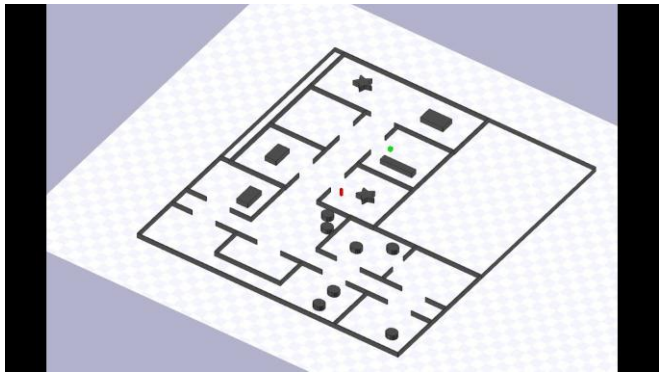
- Stage
- Demo
- Primjeri
- Priprema za iduće vježbe

crta

Stage

- Lagan i jednostavan simulator





- Instalacija primjera

```
mkdir -p stage_ws/src
cd stage_ws/src
git clone https://github.com/CRTA-Lab/Stage.git
git clone https://github.com/CRTA-Lab/stage_ros2.git
rosdep update
rosdep install --from-paths ./Stage --ignore-src -r –y
rosdep install --from-paths ./stage_ros2 --ignore-src -r -y
cd ~/stage_ws
colcon build --symlink-install
```

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- Pokretanje primjera

```
ros2 launch stage_ros2 demo.launch.py world:=cave use_stamped_velocity:=false
```

- Moguće je koristiti *namespace* za samo jednog robota

```
ros2 launch stage_ros2 stage.launch.py world:=cave enforce_prefixes:=true one_tf_tree:=false
```

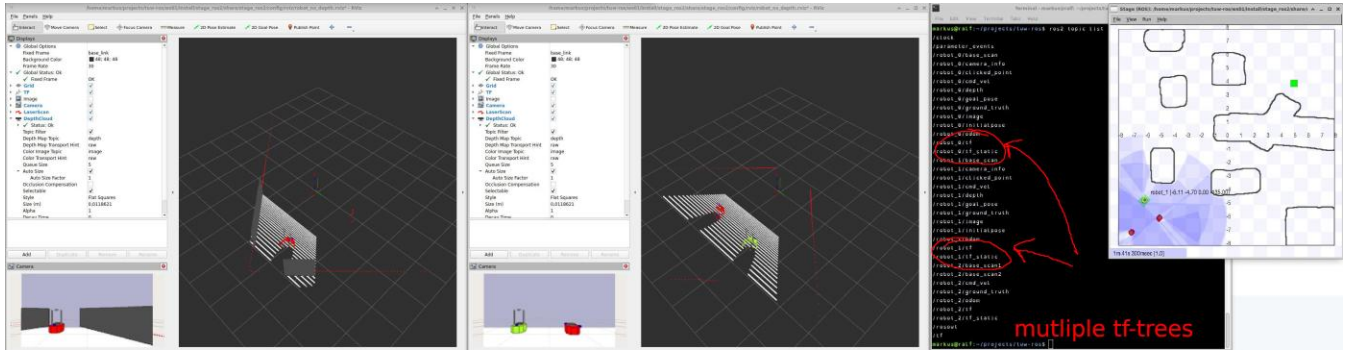
- Višerobotski sustav i više *tf*-ova

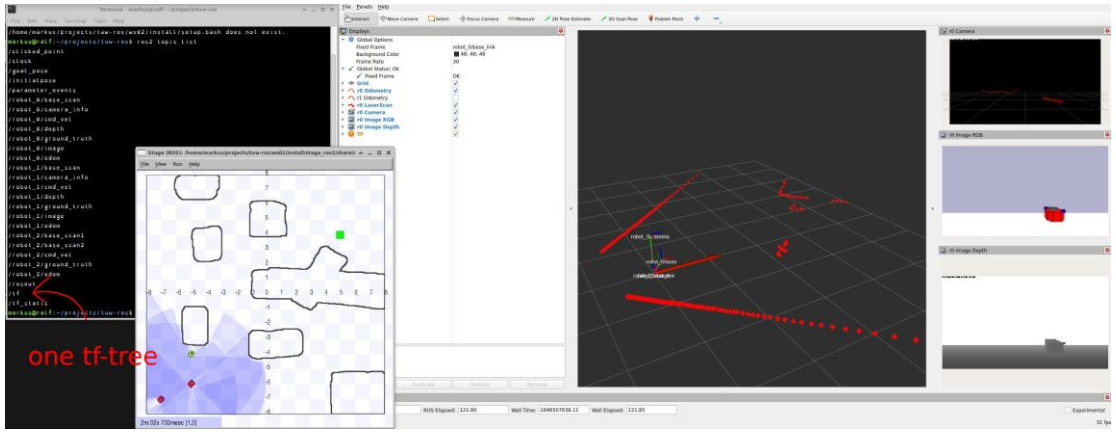
```
ros2 launch stage_ros2 stage.launch.py world:=cave_three_robots one_tf_tree:=false
ros2 launch stage_ros2 rviz_ns.launch.py config:=robot_ns namespace:=robot_0
ros2 launch stage_ros2 rviz_ns.launch.py config:=robot_ns namespace:=robot_1
```

Stage

- Višerobotski sustav i jedan *tf*-ova

```
ros2 launch stage_ros2 demo.launch.py world:=cave_three_robots one_tf_tree:=true
```





Priprema za *moveIt*

```
sudo apt install python3-vcstool

mkdir -p ~/ws_moveit2/src
cd ~/ws_moveit2/src
git clone --branch humble https://github.com/ros-planning/moveit2_tutorials
vcs import < moveit2_tutorials/moveit2_tutorials.repos
sudo apt update && rosdep install -r --from-paths . --ignore-src --rosdistro $ROS_DISTRO –y

cd ~/ws_moveit2
colcon build --mixin release
```

- Ova naredba za *build* će vjerojatno potrajati duže vrijeme (20+ minuta), ovisno o brzini vašeg računala i količini dostupne RAM memorije (preporučeno je 32 GB). Ako imate malo memorije ili općenito primjećujete da se *build* teško izvršava na vašem računalu, možete dodati argument --parallel-workers 1 uz gornju *colcon* naredbu.

Priprema za *moveIt*

- Za korištenje *moveIt*-a preporuča se koristiti *Cyclone DDS* protokol

```
sudo apt install ros-humble-rmw-cyclonedds-cpp

export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp
```

- Preporuka je da *export* naredbu dodate u *.bashrc*
- Provjera instalacije

```
ros2 launch moveit2_tutorials demo.launch.py rviz_config:=panda_moveit_config_demo_empty.rviz
```

Hvala!

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Konzultacije: Četvrtkom 12:00-13:00
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